Amiga USB-C PD Power Supply v1.0 — Amiga 500 / 600 / 1200 (and other +5/+12/−12 V machines)
Needs a USB-C PD charger that offers 15 V at 3 A (45 W or more). U1 (CH224A) requests 15 V via R1 = 56k.
U2/U4 run only while the PD contract holds (PG low -> Q1 off) and VBUS > ~11.5 V (R3/R4).
+5 V: U2 TPS56637 -> U3 eFuse (5.64 V OVLO, 6 A limit): 5 A continuous. +12 V: U4, 12.36 V. −12 V: U5 inverting, from +12 V.
Budget: 45 W in, about 40 W out, e.g. 5 V 5 A + 12 V 1 A + −12 V 0.2 A.
Original design. Spec targets only from Commodore PSU ratings and the SnCMuseum V2 board; no circuitry copied.

PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG
VBUS GND 5V_BUCK −12V +12V

J1 USB-C (power in)
U1 CH224A
U2 TPS56637RPAR
U3 TPS259470ARPWR
U4 LMR51430XDDCR
U5 LMR51430XDDCR
J2 AMIGA OUT
D1 SMF5.0A
D4 SMF13A
D6 B5819W
D7 SMF13A
D8 GREEN
LED12 GREEN
LED3 RED
LED5 GREEN
Q1 2N7002

USB-C PD 15 V -> Amiga +5 V 5 A (eFuse protected), +12 V, −12 V

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Size: A3 | Date: 2026-11-03 | **Rev: 1.0**
KiCad E.D.A. 10.0.4 | Id: 1/1